ETP story-formalization benchmark · vacuous laws excluded

Stories generated from Equational Theories Project implications (`storyform.py`), formalized back by each model over OpenRouter, and graded syntactically (`checkform.py`). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

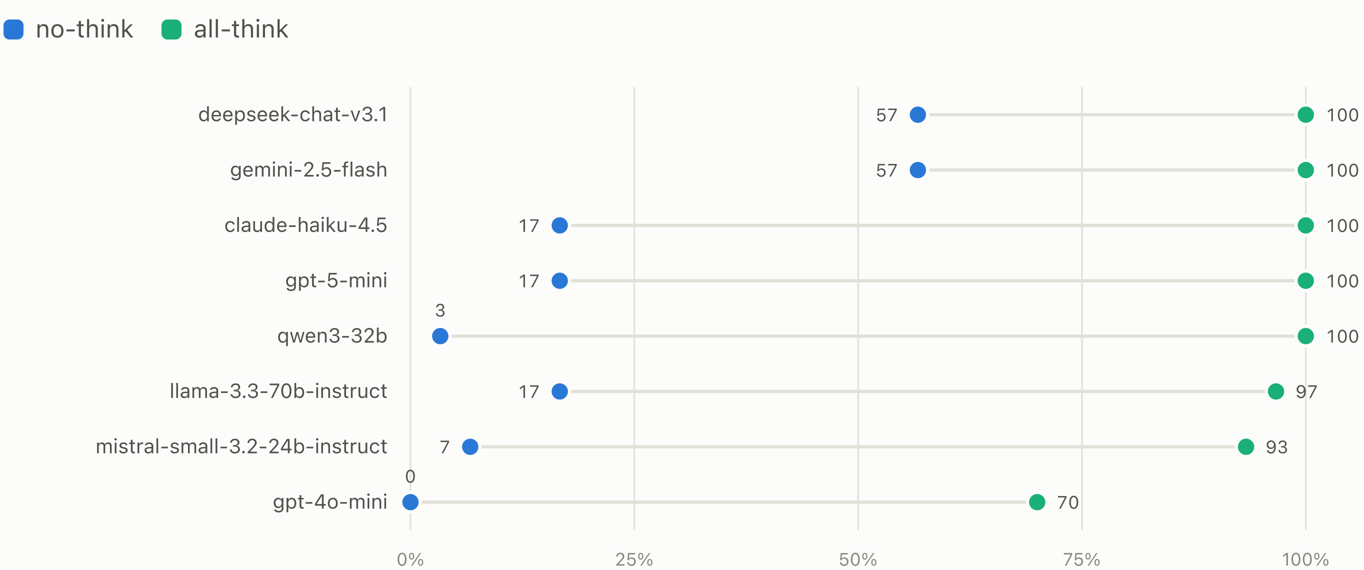
results/no-vacuous/02-reasoning-and-complexity/runs/run-s0-n30-think-off · results/no-vacuous/02-reasoning-and-complexity/runs/run-s0-n30-think-on · results/no-vacuous/02-reasoning-and-complexity/runs/run-strat5-s0-think-off · results/no-vacuous/02-reasoning-and-complexity/runs/run-strat5-s0-think-on

Runs	Models	Stories	Graded calls	Overall correct
4	8	54	864	63%

30 uniformly sampled pairs · seed 0 · no-think vs all-think

FIGURE 1 **Correct rate by model: no-think vs all-think — uniform 30**

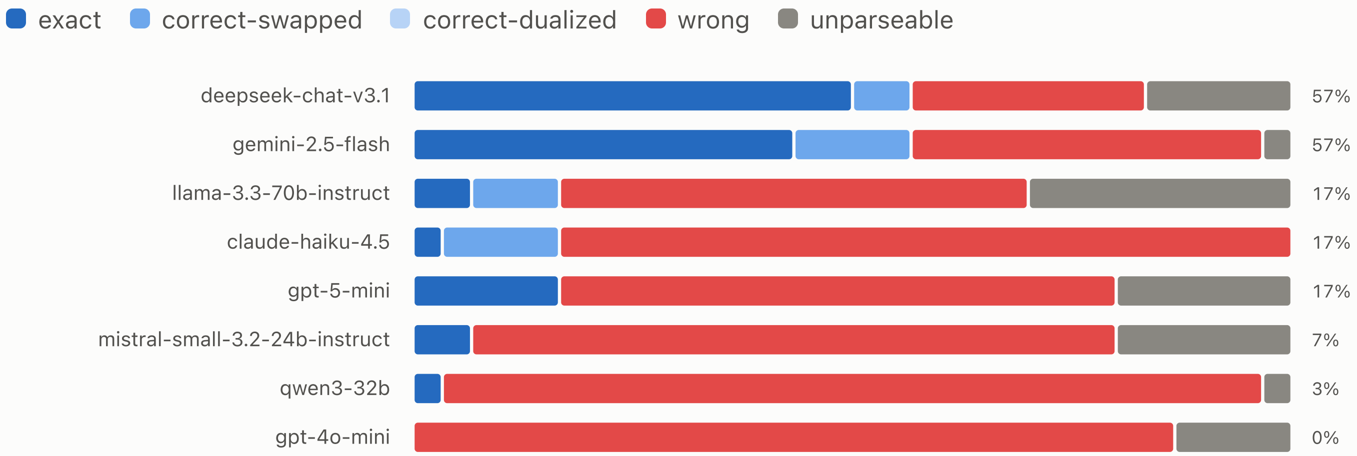
Same 30 stories under both regimes; each row connects a model's cold-pass rate to its deliberate rate. Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.



▶ Data: correct rate per model and regime

FIGURE 2 **Verdict composition — no-think regime · uniform 30**

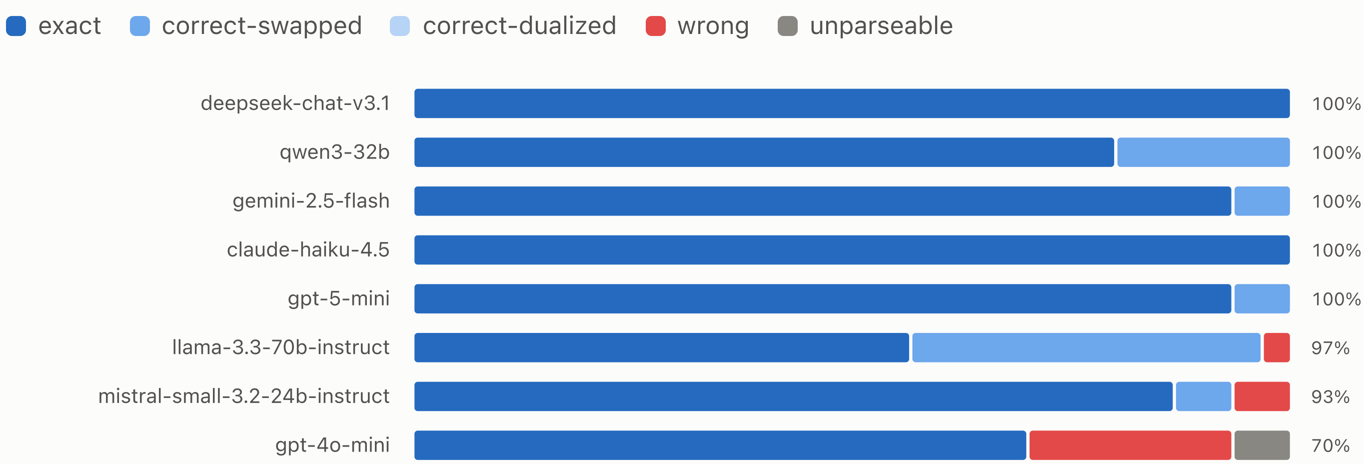
How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.



▶ Data: verdict counts per model

FIGURE 3 **Verdict composition — all-think regime · uniform 30**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.

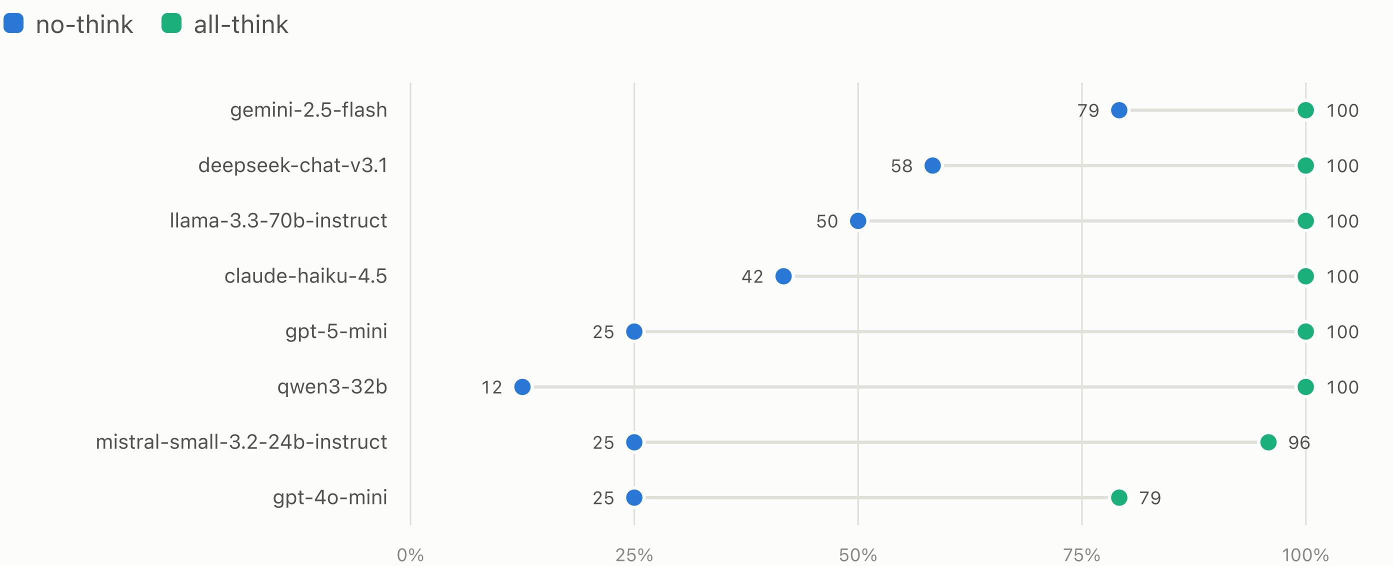


▶ Data: verdict counts per model

24 pairs, 5 per operation-count bin 1–8 · seed 0 · no-think vs all-think

FIGURE 4 **Correct rate by model: no-think vs all-think — stratified 24**

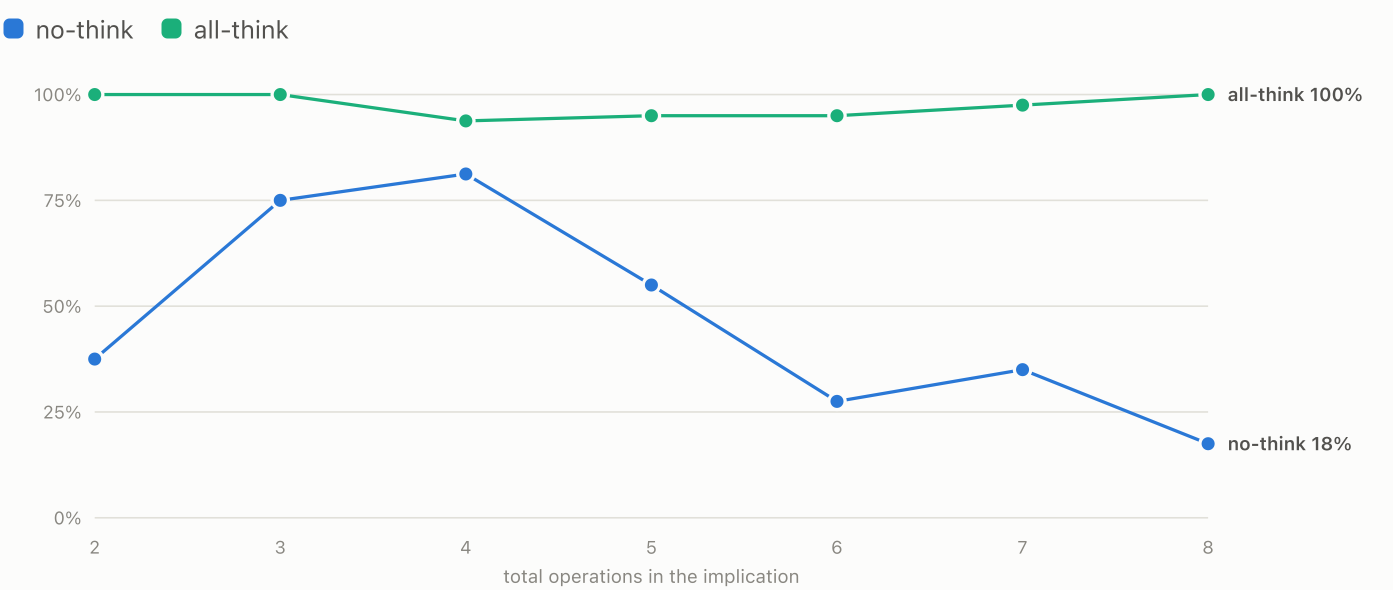
Same 24 stories under both regimes; each row connects a model's cold-pass rate to its deliberate rate. Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



▶ Data: correct rate per model and regime

FIGURE 5 **Accuracy vs operation count — stratified 24**

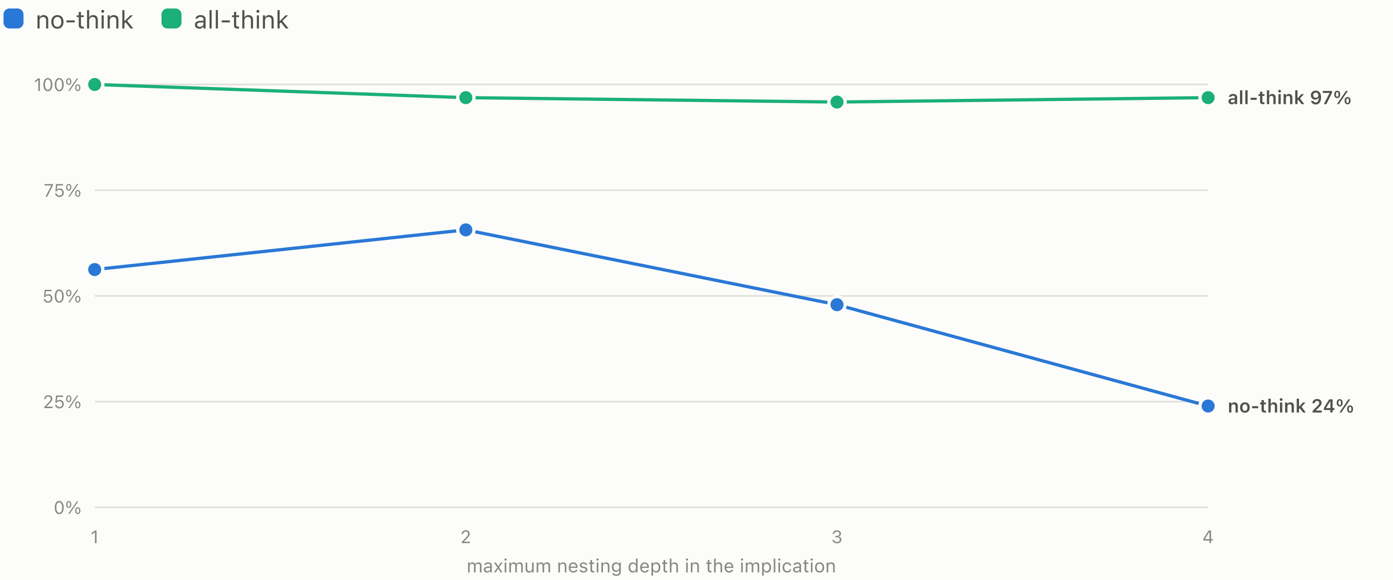
Correct rate pooled over all models, by the implication's total operation count (both equations combined). Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



▶ Data: pooled correct rate per operation-count bin

FIGURE 6 **Accuracy vs nesting depth — stratified 24**

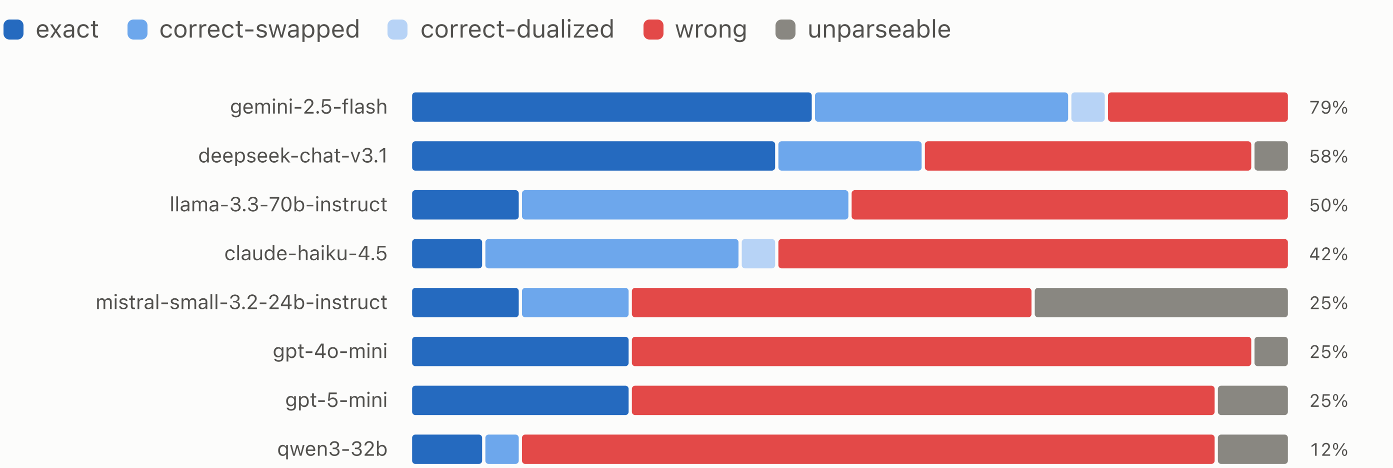
Correct rate pooled over all models, by the deepest operation nesting across the four equation sides. Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



▶ Data: pooled correct rate per depth bin

FIGURE 7 **Verdict composition — no-think regime · stratified 24**

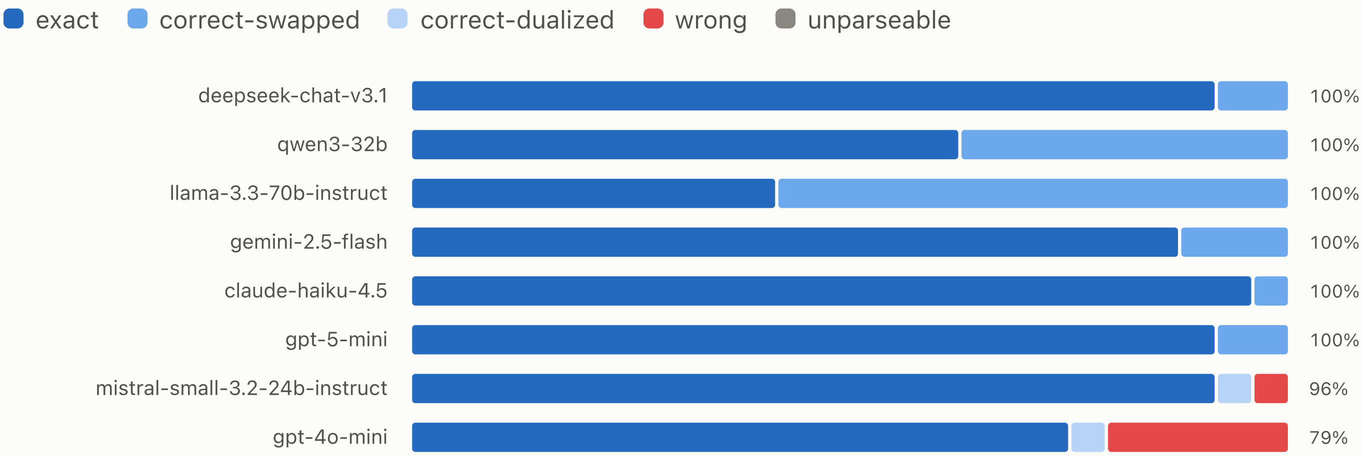
How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



▶ Data: verdict counts per model

FIGURE 8 **Verdict composition — all-think regime · stratified 24**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



▶ Data: verdict counts per model